



Seara Data

Seara, Menemani Perjalanan  
Datamu Lebih Terarah.

# Portfolio Bootcamp Data Analyst – Microsoft Excel

Batch 2

By Rahmanida S

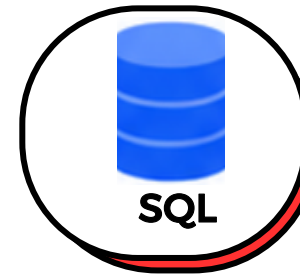
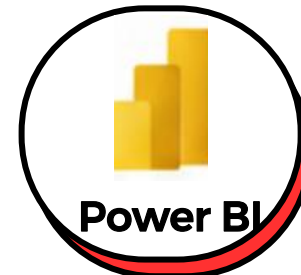




**Seara, Menemani Perjalanan  
Datamu Lebih Terarah.**

# Bootcamp Data Analyst

**Tools Wajib Data  
Analyst**



**4 Weeks Intensive Bootcamp**

## About Me

As a Chemistry graduate and an experienced QA/QC Supervisor with a four-year track record in manufacturing, I focus on driving process optimization and problem-solving. In today's landscape, I observe that the most impactful operational decisions are entirely data-driven. This insight inspired me to develop expertise in data analytics. To leverage this expertise and expand my career reach toward strategic analytical roles, I am committed to building new technical capabilities through this Data Analyst Bootcamp.

Positioning myself for a strategic transition into data analytics, I am continuously building technical skills in advanced data processing tools. I aim to merge my industrial experience with these new frameworks to architect smarter, high-impact business decisions.



**Rahmanida S**

**PESERTA BOOTCAMP**

**BOOTCAMP DATA ANALYST  
BY SEARA DATA (BATCH 2)**



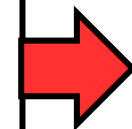
# Data Analyst Learning Journey

Seara, Menemani Perjalanan  
Datamu Lebih Terarah.

## Excel

Understand & Prepare Data  
Become a Data Analyst  
Microsoft Excel Basic  
Microsoft Excel Intermediate  
Power Query

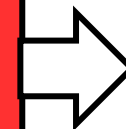
Output: Clean dataset ready for analysis.



## Power BI

Analyze & Visualize Data  
Microsoft Power BI Basic  
DAX & Data Calculation  
Microsoft Power BI Advanced  
Visualization & Interactive Report

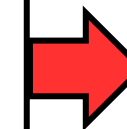
Output: Interactive dashboard and business insights.



## Python

Transform & Automate Data  
Python Basic  
Data Cleaning & Formatting  
Python Advanced  
Data Transformation

Output: Processed data pipeline and efficient workflows.



## SQL

Query & Extract Insights  
SQL Basic  
Query & Filter Data  
SQL Intermediate  
Join, Aggregation & Subquery

Output: Insights directly from databases.

# Summary of Introduction to Data Analytics & Ms. Excel

## What I Learned-Day 1

### Data Fundamentals

Learned 5 steps of the data analysis workflow :

- Define the question
- Collect the data.
- Clean the data
- Analyze the data
- Visualize and share

### Data Cleansing & Processing

- Removes extra text with TRIM.
- Replaces specific text characters using SUBSTITUTE.
- Counts total text characters using LEN
- Standards text capitalization format using PROPER/UPPER

### Logic & Aggregation

- Evaluates logical conditional testing using IF/NESTED IF/IFS
- Calculated total values based on criteria with SUMIF.
- Counts total occurrences based on criteria using COUNTIF
- Calculates average values based on criteria using AVERAGEIF.





Seara, Menemani Perjalanan  
Datamu Lebih Terarah.

# Summary of Introduction to Data Analytics & Ms. Excel

## Day 1 : Cleansing & Processing Data

| CIF     | Customer Name  | Usia | Jenis Kelamin | SND   | Kode Transaksi | Range Usia | Region            | Balance   | Month       | No Telepon  |
|---------|----------------|------|---------------|-------|----------------|------------|-------------------|-----------|-------------|-------------|
| 1173603 | Erna Susanto   | 30   | Perempuan     | SND04 | 521-17187      | 25-35      | Jawa Timur        | 7,349,830 | January-25  | 81134566    |
| 1204998 | Indah Susanti  | 25   | Perempuan     | SND12 | 808-40420      | 20-25      | Sulawesi Tenggara | 2,819,899 | February-25 | 82134527811 |
| 1285551 | Gilang Wibowo  | 49   | Laki-laki     | SND12 | 620-47433      | 35-50      | Sulawesi Tenggara | 5,489,398 | March-25    | 811322674   |
| 1299582 | Putri Hidayat  | 26   | Laki-laki     | SND05 | 413-91280      | 25-35      | Kalimantan        | 6,481,487 | April-25    | 8.58233E+12 |
| 1299582 | Putri Hidayat  | 26   | Laki-laki     | SND05 | 413-91280      | 25-35      | Kalimantan        | 6,481,487 | May-25      | 8213456789  |
| 1336094 | Intan Susanto  | 28   | Laki-laki     | SND09 | 521-60562      | 25-35      | Bali              | 9,601,023 | June-25     | 8221907561  |
| 1357884 | patrio satrio  | 21   | Laki-laki     | SND08 | 712-55086      | 20-25      | Papua             | 7,966,406 | July-25     | 81134567888 |
| 1425629 | Kartika Wijaya | 57   | Laki-laki     | SND12 | 521-81560      | >50        | Sulawesi Tenggara | 8,348,293 | August-25   | 8.08342E+12 |

### Data Definition & Collection :

- Define funding performance across regions by total and average.
- Collect gender demographics to track customer contribution trends.
- Assess data validity to count valid vs. invalid phone numbers.
- Aggregate key insights as the baseline for business decisions.



# Summary of Introduction to Data Analytics & Ms. Excel

## Day 1 : Cleansing & Processing Data

| CIF     | Customer Name  | Usia | Jenis Kelamin | SND   | Kode Transaksi | Range Usia | Region            | Balance   | Month       | No Telepon  |
|---------|----------------|------|---------------|-------|----------------|------------|-------------------|-----------|-------------|-------------|
| 1173603 | Erna Susanto   | 30   | Perempuan     | SND04 | 521-17187      | 25-35      | Jawa Timur        | 7,349,830 | January-25  | 81134566    |
| 1204998 | Indah Susanti  | 25   | Perempuan     | SND12 | 808-40420      | 20-25      | Sulawesi Tenggara | 2,819,899 | February-25 | 82134527811 |
| 1285551 | Gilang Wibowo  | 49   | Laki-laki     | SND12 | 620-47433      | 35-50      | Sulawesi Tenggara | 5,489,398 | March-25    | 811322674   |
| 1299582 | Putri Hidayat  | 26   | Laki-laki     | SND05 | 413-91280      | 25-35      | Kalimantan        | 6,481,487 | April-25    | 8.58233E+12 |
| 1299582 | Putri Hidayat  | 26   | Laki-laki     | SND05 | 413-91280      | 25-35      | Kalimantan        | 6,481,487 | May-25      | 8213456789  |
| 1336094 | Intan Susanto  | 28   | Laki-laki     | SND09 | 521-60562      | 25-35      | Bali              | 9,601,023 | June-25     | 8221907561  |
| 1357884 | patrio satrio  | 21   | Laki-laki     | SND08 | 712-55086      | 20-25      | Papua             | 7,966,406 | July-25     | 81134567888 |
| 1425629 | Kartika Wijaya | 57   | Laki-laki     | SND12 | 521-81560      | >50        | Sulawesi Tenggara | 8,348,293 | August-25   | 8.08342E+12 |

### Identified Problems Raw Data :

1. Scientific Notation ('8.5823')
2. Missing Leading Zeros
3. Invisible spaces
4. Text Inconsistency.

### Data Cleansing and Processing Solutions :

1. Text formatting fixed : Prepend standard "0" prefix
2. Standarize text capitalization using PROPER.
3. Clean extra spaces using TRIM
4. Count text character digits automatically using LEN
5. Evaluate logical validation status using IF or IFS .



Seara, Menemani Perjalanan  
Datamu Lebih Terarah.

# Summary of Introduction to Data Analytics & Ms. Excel

## Day 1 : Cleansing & Processing Data

| Customer Name  | Usia | Jenis Kelamin | SND   | Kode Transaksi | Range Usia | Region            | Balance   | Month       | No Telepon     | Proper         | Digit Number | Flag    |
|----------------|------|---------------|-------|----------------|------------|-------------------|-----------|-------------|----------------|----------------|--------------|---------|
| Erna Susanto   | 30   | Perempuan     | SND04 | 521-17187      | 25-35      | Jawa Timur        | 7,349,830 | January-25  | 081134566      | Erna Susanto   | 8            | Valid   |
| Indah Susanti  | 25   | Perempuan     | SND12 | 808-40420      | 20-25      | Sulawesi Tenggara | 2,819,899 | February-25 | 082134527811   | Indah Susanti  | 11           | Valid   |
| Gilang Wibowo  | 49   | Laki-laki     | SND12 | 620-47433      | 35-50      | Sulawesi Tenggara | 5,489,398 | March-25    | 0811322674     | Gilang Wibowo  | 9            | Valid   |
| Putri Hidayat  | 26   | Laki-laki     | SND05 | 413-91280      | 25-35      | Kalimantan        | 6,481,487 | April-25    | 08582334567896 | Putri Hidayat  | 13           | Invalid |
| Putri Hidayat  | 26   | Laki-laki     | SND05 | 413-91280      | 25-35      | Kalimantan        | 6,481,487 | May-25      | 08213456789    | Putri Hidayat  | 10           | Valid   |
| Intan Susanto  | 28   | Laki-laki     | SND09 | 521-60562      | 25-35      | Bali              | 9,601,023 | June-25     | 08221907561    | Intan Susanto  | 10           | Valid   |
| patrio satrio  | 21   | Laki-laki     | SND08 | 712-55086      | 20-25      | Papua             | 7,966,406 | July-25     | 081134567888   | Patrio Satrio  | 11           | Valid   |
| Kartika Wijaya | 57   | Laki-laki     | SND12 | 521-81560      | >50        | Sulawesi Tenggara | 8,348,293 | August-25   | 08083421567892 | Kartika Wijaya | 13           | Invalid |

### Result of Cleansing & Processing Data :

- All scientific notations were completely resolved, and the standard leading zero prefix was successfully restored.
- Inconsistent text formatting and capitalization issues on customer names were fully cleaned and normalized.
- Successfully separated valid phone numbers from invalid ones





Seara, Menemani Perjalanan  
Datamu Lebih Terarah.

# Summary of Introduction to Data Analytics & Ms. Excel

## Day 1 : Cleansing & Processing Data

| Region             | Total Funding      |                    |                    | Average Funding   |                   |                    |
|--------------------|--------------------|--------------------|--------------------|-------------------|-------------------|--------------------|
|                    | Laki-laki          | Perempuan          | Total              | Laki-laki         | Perempuan         | Average Total      |
| Unmapping          | 23,509,134         | 30,587,651         | 54,096,785         | 4,701,827         | 7,646,913         | 12,348,740         |
| Sulawesi Tenggara  | 38,889,087         | 32,526,914         | 71,416,001         | 6,481,515         | 5,421,152         | 11,902,667         |
| Sumatera Utara     | 6,068,646          | 35,952,769         | 42,021,415         | 6,068,646         | 5,136,110         | 11,204,756         |
| Kalimantan         | 26,694,161         | 36,954,929         | 63,649,090         | 5,338,832         | 5,279,276         | 10,618,108         |
| Sulawesi Utara     | 46,882,279         | 33,191,143         | 80,073,422         | 5,860,285         | 4,741,592         | 10,601,877         |
| Jawa Tengah        | 32,209,403         | 57,906,707         | 90,116,110         | 4,601,343         | 5,790,671         | 10,392,014         |
| DKI Jakarta        | 43,733,943         | 11,610,629         | 55,344,572         | 6,247,706         | 3,870,210         | 10,117,916         |
| Jawa Timur         | 53,113,216         | 40,757,454         | 93,870,670         | 4,828,474         | 5,094,682         | 9,923,156          |
| Bali               | 65,245,718         | 22,095,145         | 87,340,863         | 6,524,572         | 3,156,449         | 9,681,021          |
| Sumatera Selatan   | 45,059,950         | 22,388,305         | 67,448,255         | 5,006,661         | 4,477,661         | 9,484,322          |
| Papua              | 28,600,099         | 17,977,450         | 46,577,549         | 5,720,020         | 3,595,490         | 9,315,510          |
| Jawa Barat         | 31,828,143         | 44,774,317         | 76,602,460         | 3,978,518         | 4,974,924         | 8,953,442          |
| Maluku             | 20,888,685         | 29,937,365         | 50,826,050         | 4,177,737         | 3,742,171         | 7,919,908          |
| <b>Grand Total</b> | <b>462,722,464</b> | <b>416,660,778</b> | <b>879,383,242</b> | <b>69,536,136</b> | <b>62,927,300</b> | <b>132,463,435</b> |

### Business Aggregation & Key Insights:

- **Top Performing Region (Average):**  
Unmapping recorded the highest average funding at IDR 12.35 Million, followed closely by Sulawesi Tenggara at IDR 11.90 Million.
- **Lowest Performing Region (Average):**  
Maluku recorded the lowest average funding at IDR 7.92 Million.
- **Gender Contribution Trend:** Across most regions, Male customers consistently maintained a higher average funding (IDR 69.54 Million) compared to Female customers (IDR 62.93 Million).



# Summary of Introduction to Data Analytics & Ms. Excel

## Day 1 : Cleansing & Processing Data

| Region            | Number Identification |         |              | Percentage of Number Identification |           |
|-------------------|-----------------------|---------|--------------|-------------------------------------|-----------|
|                   | Valid                 | Invalid | Total Number | % Valid                             | % Invalid |
| Bali              | 14                    | 3       | 17           | 82%                                 | 18%       |
| DKI Jakarta       | 8                     | 2       | 10           | 80%                                 | 20%       |
| Jawa Barat        | 12                    | 5       | 17           | 71%                                 | 29%       |
| Jawa Tengah       | 13                    | 4       | 17           | 76%                                 | 24%       |
| Jawa Timur        | 14                    | 5       | 19           | 74%                                 | 26%       |
| Kalimantan        | 8                     | 4       | 12           | 67%                                 | 33%       |
| Maluku            | 11                    | 2       | 13           | 85%                                 | 15%       |
| Papua             | 7                     | 3       | 10           | 70%                                 | 30%       |
| Sulawesi Tenggara | 9                     | 3       | 12           | 75%                                 | 25%       |
| Sulawesi Utara    | 11                    | 4       | 15           | 73%                                 | 27%       |
| Sumatera Selatan  | 10                    | 4       | 14           | 71%                                 | 29%       |
| Sumatera Utara    | 6                     | 2       | 8            | 75%                                 | 25%       |
| Unmapping         | 5                     | 4       | 9            | 56%                                 | 44%       |
| Grand Total       | 128                   | 45      | 173          | 74%                                 | 26%       |

### Business Aggregation & Key Insights:

- **Data Reliability:** 74% of total phone numbers is **Valid**, ensuring the overall analysis is highly reliable.
- **Low Risk Regions:** **Maluku (85%)** and **Bali (82%)** has the cleanest data, making them ideal for upcoming funding campaigns.
- **High Risk Regions:** **Unmapping (44%)** and **Kalimantan (33%)** have the highest invalid numbers, requiring immediate data cleansing despite their high funding potential.



# Summary of Advance Function In Microsoft Excel

## What I Learned-Day 2

### Advance Excel Functions

Learned core data retrieval & logic tools :

- Mastered lookup methods with VLOOKUP and XLOOKUP.
- Combined INDEX & MATCH for flexible dynamic lookups.
- Pivot Table for quick and flexible data aggregation

### Data Visualization Principles

Mastered effective presentation rules :

- Ensures Clarity & Simplicity by avoiding 3D charts and excessive colors.
- Provides critical context using clear titles, subtitles, and units
- Selects the right chart type such as Line, Bar, or Stacked Chart

### Data Driven Decision Making

Translates insights into strategic business actions:

- Identifies trend patterns, peak performance, and anomalies.
- Investigates key drivers behind performance changes.



Seara, Menemani Perjalanan  
Datamu Lebih Terarah.

# Summary of Advance Function In Microsoft Excel

## Day 2 : Data Cleansing & Joining

| CIF     | Cust_Name     | Usia | Jenis Kelamin | Date_Cif_Open | Acc_Number | Date_Acc_Open | Product_Cod | Balance      | Avg_Balance  | SND   | Region | Product Name | Cust Segment | Cust Profile | Flag |
|---------|---------------|------|---------------|---------------|------------|---------------|-------------|--------------|--------------|-------|--------|--------------|--------------|--------------|------|
| 2287021 | Budi Kusuma   | 35   | Perempuan     | 5/21/2021     | 445036     | 5/31/2021     | 621         | 2,348,116.00 | 8,955,367.00 | SND01 |        |              |              |              |      |
| 3684345 | Wulan Wibowo  | 29   | Laki-laki     | 5/5/2025      | 690376     | 5/15/2025     | 620         | 4,632,122.00 | 6,250,736.00 | SND02 |        |              |              |              |      |
| 6507231 | Yudi Santoso  | 37   | Perempuan     | 7/19/2022     | 726795     | 7/29/2022     | 521         | 4,676,007.00 | 6,844,373.00 | SND03 |        |              |              |              |      |
| 2306909 | Hendra Permar | 25   | Perempuan     | 12/17/2024    | 471359     | 12/27/2024    | 808         | 5,717,876.00 | 8,325,624.00 | SND04 |        |              |              |              |      |
| 6713678 | Rina Rahayu   | 29   | Perempuan     | 3/11/2022     | 222391     | 3/21/2022     | 413         | 2,158,895.00 | 8,347,943.00 | SND05 |        |              |              |              |      |
| 3477250 | Lina Wibowo   | 45   | Perempuan     | 3/7/2023      | 784959     | 3/17/2023     | 712         | 2,203,171.00 | 5,618,396.00 | SND06 |        |              |              |              |      |
| 2446706 | Joko Sari     | 55   | Perempuan     | 12/31/2020    | 670347     | 1/10/2021     | 621         | 3,722,178.00 | 6,563,564.00 | SND07 |        |              |              |              |      |

| Cust Segmen | Tier Prioritas |         |         |         |
|-------------|----------------|---------|---------|---------|
|             | Grade A        | Grade B | Grade C | Grade D |
| Mass Market | Low            | Low     | Low     | Low     |
| Optimal     | High           | High    | Middle  | Middle  |
| Privilege   | High           | High    | High    | High    |
| Segment     | Balance        | Tier    | Usia    |         |
| Mass Market | < 10 Juta      | Grade A | 25-35   |         |
| Optimal     | 10-50 Juta     | Grade B | 35-55   |         |
| Privilege   | > 50 Juta      | Grade C | < 25    |         |
|             |                | Grade D | > 55    |         |

### Data Definition & Collection :

- Define marketing growth strategy by transforming raw customer datasets into highly targeted insights
- Collect and analyze customer profiles to generate high-potential leads based on corporate criteria.
- Visualize data analytics trends to deliver actionable frameworks for banking product expansion.



Seara, Menemani Perjalanan  
Datamu Lebih Terarah.

# Summary of Advance Function In Microsoft Excel

## Day 2 : Data Cleansing & Joining

| CIF     | Cust_Name     | Usia | Jenis Kelamin | Date_Cif_Open | Acc_Number | Date_Acc_Open | Product_Cod | Balance      | Avg_Balance  | SND   | Region | Product Name | Cust Segment | Cust Profile | Flag |
|---------|---------------|------|---------------|---------------|------------|---------------|-------------|--------------|--------------|-------|--------|--------------|--------------|--------------|------|
| 2287021 | Budi Kusuma   | 35   | Perempuan     | 5/21/2021     | 445036     | 5/31/2021     | 621         | 2,348,116.00 | 8,955,367.00 | SND01 |        |              |              |              |      |
| 3684345 | Wulan Wibowo  | 29   | Laki-laki     | 5/5/2025      | 690376     | 5/15/2025     | 620         | 4,632,122.00 | 6,250,736.00 | SND02 |        |              |              |              |      |
| 6507231 | Yudi Santoso  | 37   | Perempuan     | 7/19/2022     | 726795     | 7/29/2022     | 521         | 4,676,007.00 | 6,844,373.00 | SND03 |        |              |              |              |      |
| 2306909 | Hendra Permar | 25   | Perempuan     | 12/17/2024    | 471359     | 12/27/2024    | 808         | 5,717,876.00 | 8,325,624.00 | SND04 |        |              |              |              |      |
| 6713678 | Rina Rahayu   | 29   | Perempuan     | 3/11/2022     | 222391     | 3/21/2022     | 413         | 2,158,895.00 | 8,347,943.00 | SND05 |        |              |              |              |      |
| 3477250 | Lina Wibowo   | 45   | Perempuan     | 3/7/2023      | 784959     | 3/17/2023     | 712         | 2,203,171.00 | 5,618,396.00 | SND06 |        |              |              |              |      |
| 2446706 | Joko Sari     | 55   | Perempuan     | 12/31/2020    | 670347     | 1/10/2021     | 621         | 3,722,178.00 | 6,563,564.00 | SND07 |        |              |              |              |      |

### Identified Problems Raw Data :

1. Unlinked Area Codes
2. Hidden Product Name
3. Granular Raw Balances
4. Uncategorized Demographics
5. No Lead Priority

### Data Cleansing & Joining Solutions :

1. Map region data using XLOOKUP
2. Extract product name using XLOOKUP/VLOOKUP
3. Segment customer balances using IFS
4. Profile age demographics using IFS.
5. Evaluate lead priority using INDEX & MATCH.





Seara, Menemani Perjalanan  
Datamu Lebih Terarah.

# Summary of Advance Function In Microsoft Excel

## Day 2 : Data Cleansing & Joining

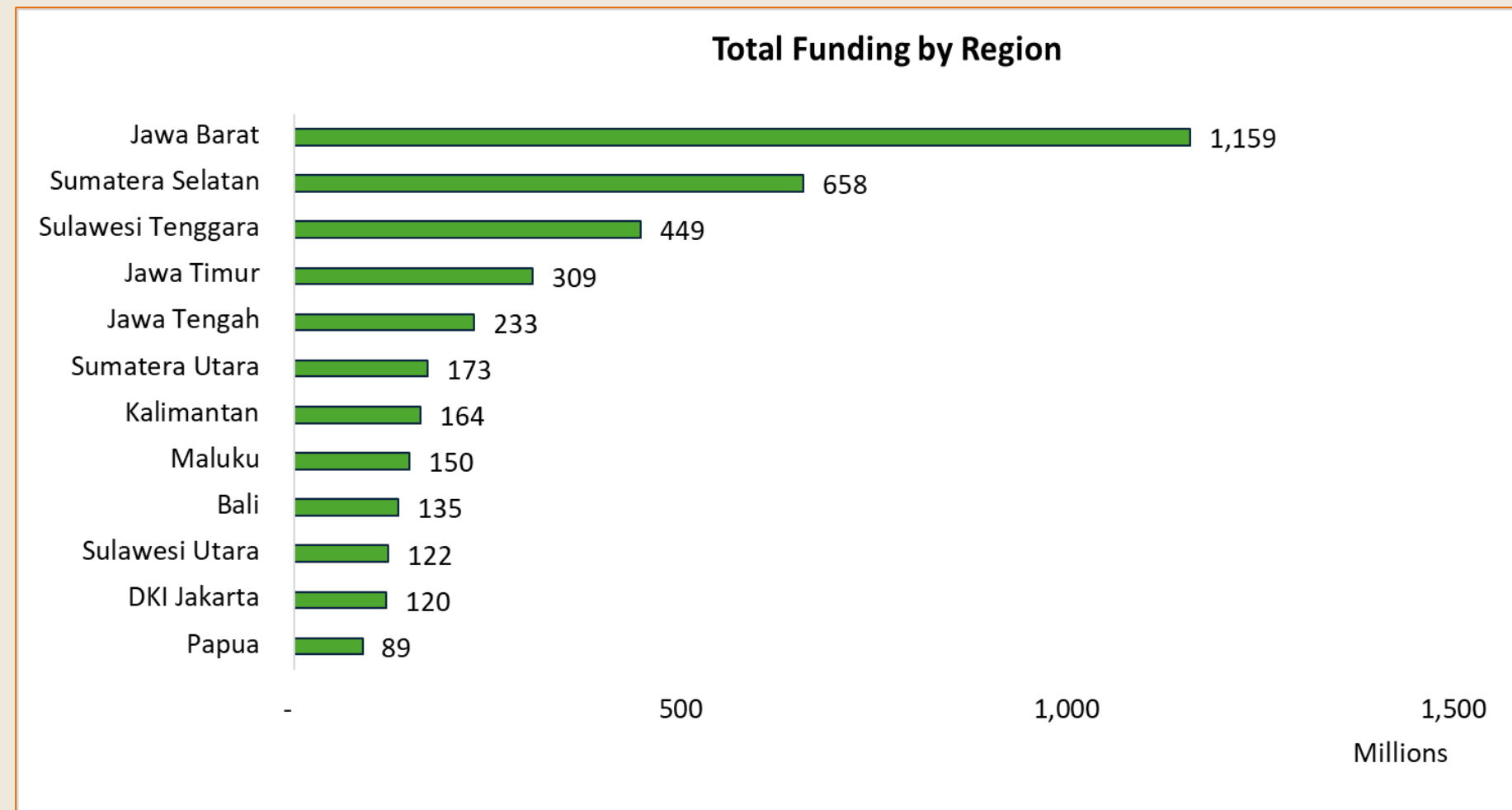
| Cust_Name         | Usia | Jenis Kelam | Product_Cod | Balance       | Avg_Balance   | SND   | Region           | Product Name | Cust Segmen | Cust Profiling | Flag |
|-------------------|------|-------------|-------------|---------------|---------------|-------|------------------|--------------|-------------|----------------|------|
| Budi Kusuma       | 35   | Perempuan   | 621         | 2,348,116.00  | 8,955,367.00  | SND01 | Jawa Barat       | Tab Hajj     | Mass Market | Grade A        | Low  |
| Wulan Wibowo      | 29   | Laki-laki   | 620         | 4,632,122.00  | 6,250,736.00  | SND02 | Sumatera Selatan | Wadiah       | Mass Market | Grade A        | Low  |
| Yudi Santoso      | 37   | Perempuan   | 521         | 4,676,007.00  | 6,844,373.00  | SND03 | Jawa Tengah      | Special Hajj | Mass Market | Grade B        | Low  |
| Hendra Permana    | 25   | Perempuan   | 808         | 5,717,876.00  | 8,325,624.00  | SND04 | Jawa Timur       | Tab Umroh    | Mass Market | Grade A        | Low  |
| Rina Rahayu       | 29   | Perempuan   | 413         | 2,158,895.00  | 8,347,943.00  | SND05 | Kalimantan       | Tab Valas    | Mass Market | Grade A        | Low  |
| Lina Wibowo       | 45   | Perempuan   | 712         | 2,203,171.00  | 5,618,396.00  | SND06 | Sumatera Utara   | Tab Syariah  | Mass Market | Grade B        | Low  |
| Joko Sari         | 55   | Perempuan   | 621         | 3,722,178.00  | 6,563,564.00  | SND07 | DKI Jakarta      | Tab Hajj     | Mass Market | Grade D        | Low  |
| Hadi Saputri      | 20   | Perempuan   | 620         | 2,173,515.00  | 5,182,322.00  | SND08 | Papua            | Wadiah       | Mass Market | Grade C        | Low  |
| Kartika Kurniawan | 64   | Perempuan   | 521         | 5,867,720.00  | 7,688,655.00  | SND09 | Bali             | Special Hajj | Mass Market | Grade D        | Low  |
| Indah Ramadhan    | 31   | Perempuan   | 808         | 7,629,989.00  | 6,784,112.00  | SND10 | Maluku           | Tab Umroh    | Mass Market | Grade A        | Low  |
| Dewi Nugroho      | 28   | Laki-laki   | 413         | 18,287,346.00 | 20,678,701.00 | SND11 | Sulawesi Utara   | Tab Valas    | Optimal     | Grade A        | High |

### After Cleansing & Joining Data :

- All missing fields are completed by connecting the master lookup table.
- Customers are successfully grouped based on balance and age criteria.
- Marketing priority status (flag) is clearly defined for targeted campaigns.

# Summary of Advance Function In Microsoft Excel

## Day 2 : Data Visualization



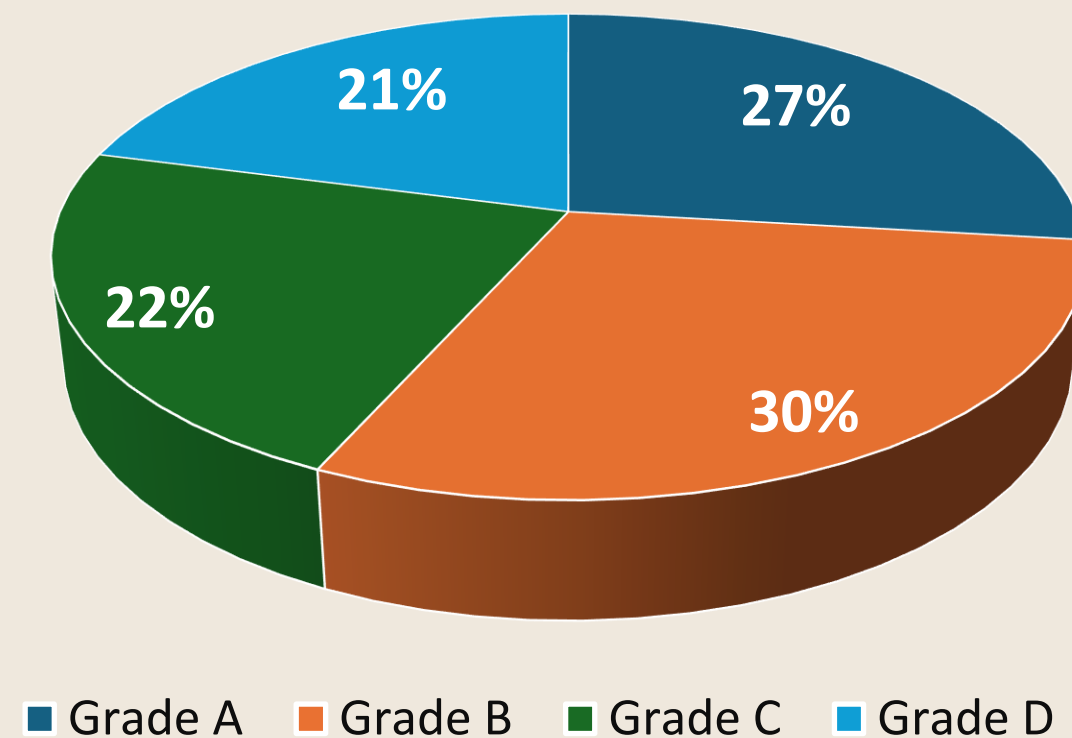
### Business Aggregation & Key Insights:

- **Top Performance:** Jawa Barat dominates regional performance, contributing the highest total funding at IDR 1,159 Million.
- **Major Regions:** Sumatera Selatan (IDR 658 Million) and Sulawesi Tenggara (IDR 449 Million) emerge as the next primary growth drivers.
- **Low Performing Areas:** Papua records the lowest funding contribution at IDR 89 Million, indicating a potential segment for targeted market development.

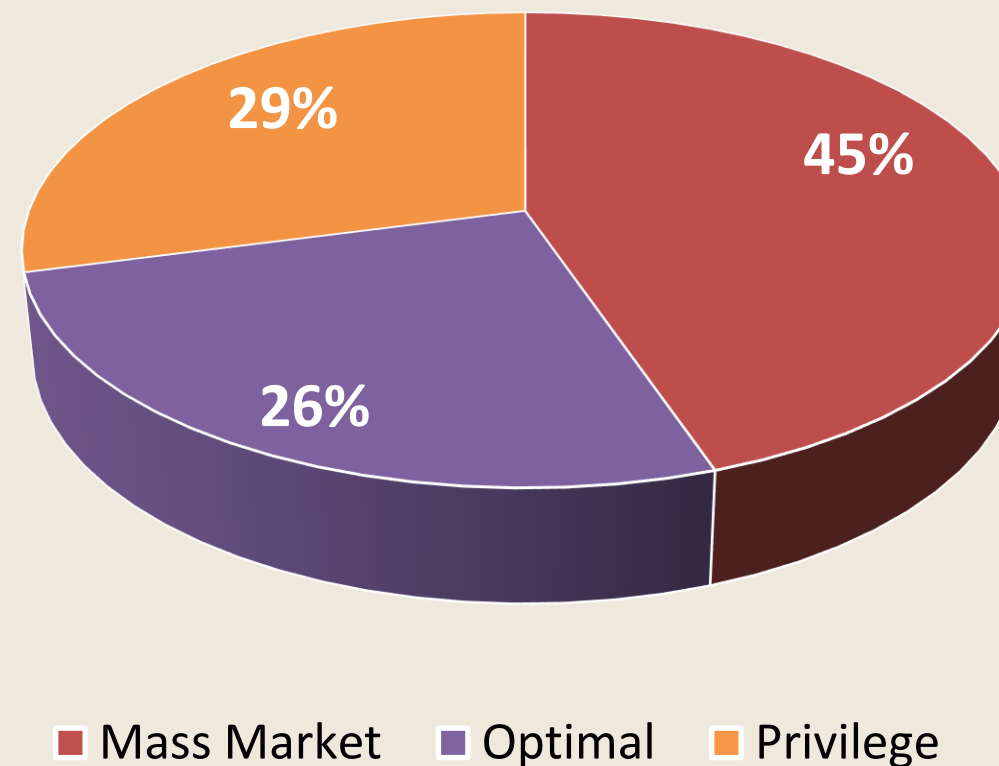
# Summary of Advance Function In Microsoft Excel

## Day 2 : Data Visualization

Customer Profiling All Region



Customer Segment All Regions



### Business Aggregation & Key Insights:

- **Age Demographics:** Grade B (30%) and Grade A (27%) lead the portfolio, showing that productive age is our main audience.
- **Financial Segmentation:** Mass Market (45%) forms the largest customer base, followed by a balanced split between Privilege (29%) and Optimal (26%) segments.
- **Strategic Takeaway:** Marketing campaigns should focus heavily on high-volume **Mass Market** products tailored for **Grade A & B** demographic tiers to maximize conversions

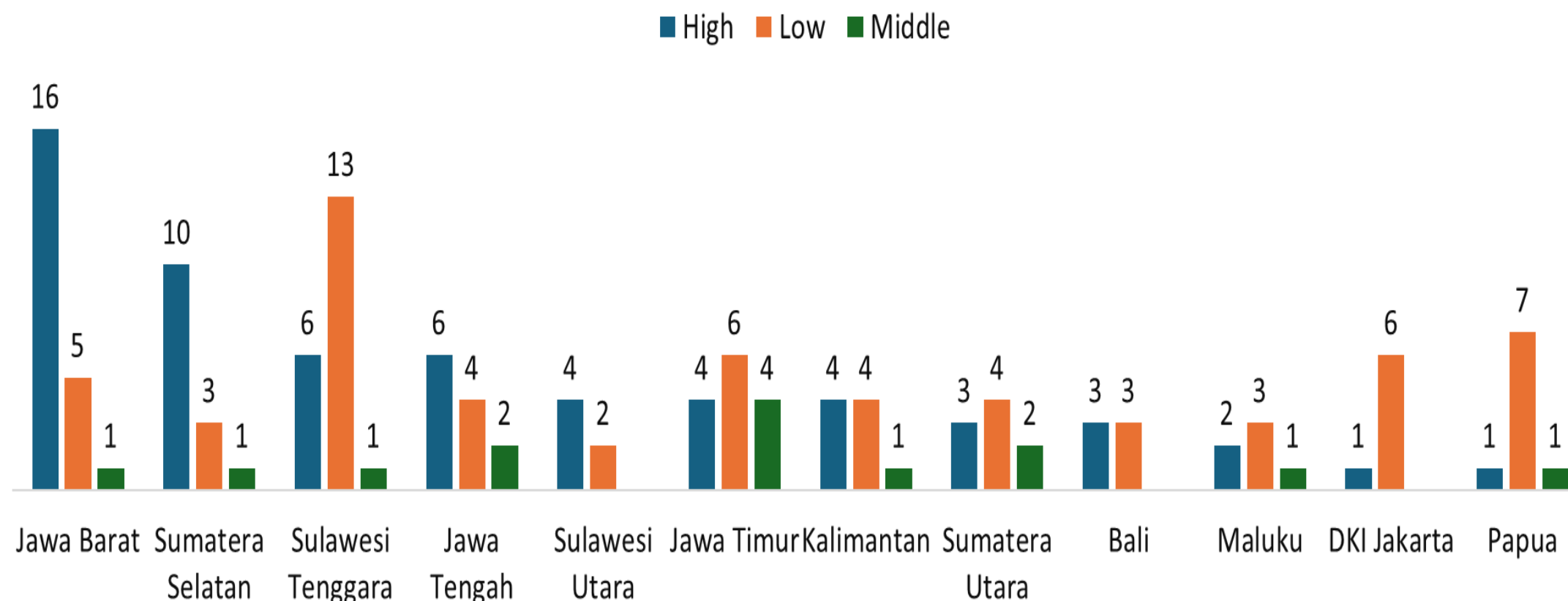
# Summary of Advance Function In Microsoft Excel

## Day 2 : Data Visualization

### Business Aggregation & Key Insights:

- **Top Regions (High):** Jawa Barat (16) and Sumatera Selatan (10) have the most High priority customers for premium campaigns.
- **Top Regions (Low):** Sulawesi Tenggara (13) holds the largest group of Low priority customers for mass marketing.
- **Equal Growth:** Bali shows an even split between high and low priority customer segments.

Tier Priority Customer By Regions



# Lampiran

- Dokumentasi
- Script



Seara, Menemani Perjalanan  
Datamu Lebih Terarah.

## 1. Documentation & Script for Joining Data using Xlookup

| =XLOOKUP(K3,Lookup!\$G\$6:\$G\$17,Lookup!\$F\$6:\$F\$17,"Tidak ada",0) |      |             |             |              |              |       |                  |              |             |                |      |
|------------------------------------------------------------------------|------|-------------|-------------|--------------|--------------|-------|------------------|--------------|-------------|----------------|------|
| B                                                                      | C    | D           | H           | I            | J            | K     | L                | M            | N           | O              | P    |
| Cust_Name                                                              | Usia | Jenis Kelam | Product_Coc | Balance      | Avg_Balance  | SND   | Region           | Product Name | Cust Segmen | Cust Profiling | Flag |
| Budi Kusuma                                                            | 35   | Perempuan   | 621         | 2,348,116.00 | 8,955,367.00 | SND01 | Jawa Barat       | Tab Hajj     | Mass Market | Grade A        | Low  |
| Wulan Wibowo                                                           | 29   | Laki-laki   | 620         | 4,632,122.00 | 6,250,736.00 | SND02 | Sumatera Selatan | Wadiah       | Mass Market | Grade A        | Low  |
| Yudi Santoso                                                           | 37   | Perempuan   | 521         | 4,676,007.00 | 6,844,373.00 | SND03 | Jawa Tengah      | Special Hajj | Mass Market | Grade B        | Low  |
| Hendra Permana                                                         | 25   | Perempuan   | 808         | 5,717,876.00 | 8,325,624.00 | SND04 | Jawa Timur       | Tab Umroh    | Mass Market | Grade A        | Low  |

## 2. Documentation & Script for Joining Data using IFS

| =IFS(I3<10000000,"Mass Market",I3<=50000000,"Optimal",I3>50000000,"Privilege") |      |             |             |              |              |       |                  |              |             |                |      |
|--------------------------------------------------------------------------------|------|-------------|-------------|--------------|--------------|-------|------------------|--------------|-------------|----------------|------|
| B                                                                              | C    | D           | H           | I            | J            | K     | L                | M            | N           | O              | P    |
| Cust_Name                                                                      | Usia | Jenis Kelam | Product_Coc | Balance      | Avg_Balance  | SND   | Region           | Product Name | Cust Segmen | Cust Profiling | Flag |
| Budi Kusuma                                                                    | 35   | Perempuan   | 621         | 2,348,116.00 | 8,955,367.00 | SND01 | Jawa Barat       | Tab Hajj     | Mass Market | Grade A        | Low  |
| Wulan Wibowo                                                                   | 29   | Laki-laki   | 620         | 4,632,122.00 | 6,250,736.00 | SND02 | Sumatera Selatan | Wadiah       | Mass Market | Grade A        | Low  |
| Yudi Santoso                                                                   | 37   | Perempuan   | 521         | 4,676,007.00 | 6,844,373.00 | SND03 | Jawa Tengah      | Special Hajj | Mass Market | Grade B        | Low  |

## 3. Documentation & Script for Joining Data using INDEX & MATCH

| =INDEX(Lookup!\$J\$7:\$M\$9,MATCH(Dataset!N3,Lookup!\$I\$7:\$I\$9,0),MATCH(Dataset!O3,Lookup!\$J\$6:\$M\$6,0)) |      |             |             |              |              |       |                  |              |             |                |      |
|----------------------------------------------------------------------------------------------------------------|------|-------------|-------------|--------------|--------------|-------|------------------|--------------|-------------|----------------|------|
| B                                                                                                              | C    | D           | H           | I            | J            | K     | L                | M            | N           | O              | P    |
| Cust_Name                                                                                                      | Usia | Jenis Kelam | Product_Coc | Balance      | Avg_Balance  | SND   | Region           | Product Name | Cust Segmen | Cust Profiling | Flag |
| Budi Kusuma                                                                                                    | 35   | Perempuan   | 621         | 2,348,116.00 | 8,955,367.00 | SND01 | Jawa Barat       | Tab Hajj     | Mass Market | Grade A        | Low  |
| Wulan Wibowo                                                                                                   | 29   | Laki-laki   | 620         | 4,632,122.00 | 6,250,736.00 | SND02 | Sumatera Selatan | Wadiah       | Mass Market | Grade A        | Low  |
| Yudi Santoso                                                                                                   | 37   | Perempuan   | 521         | 4,676,007.00 | 6,844,373.00 | SND03 | Jawa Tengah      | Special Hajj | Mass Market | Grade B        | Low  |





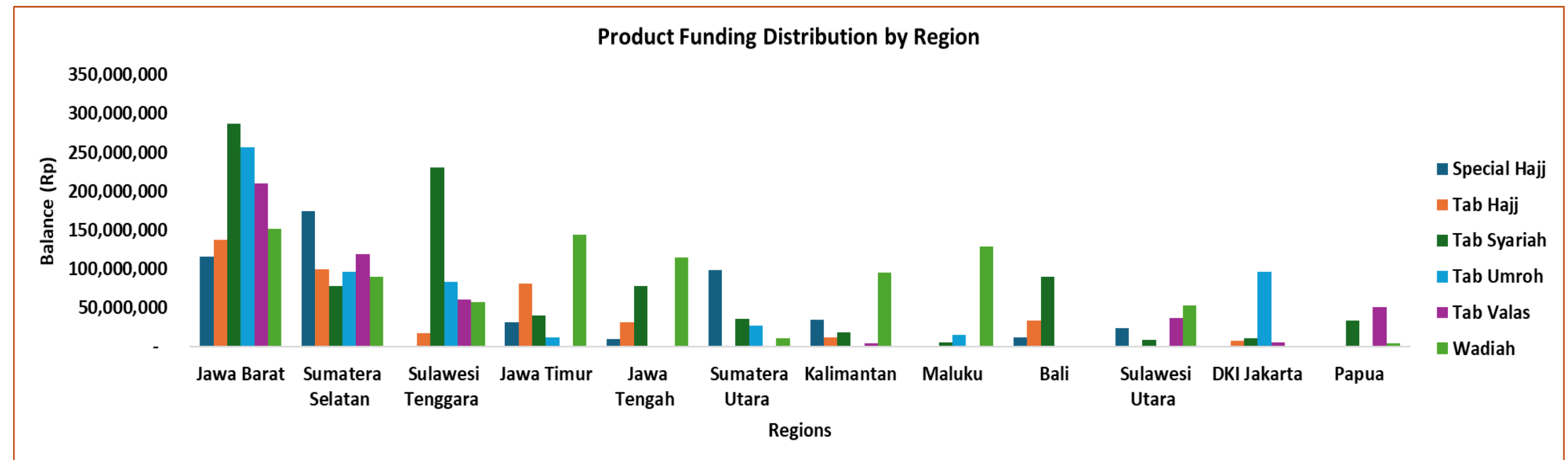
Seara, Menemani Perjalanan  
Datamu Lebih Terarah.

# Lampiran

- Dashboard
- Link Github

## 1. Dashboard Product Funding Distribution by Region

| Region            | Product Name |             |             |             |             |             | Grand Total   |
|-------------------|--------------|-------------|-------------|-------------|-------------|-------------|---------------|
|                   | Special Hajj | Tab Hajj    | Tab Syariah | Tab Umroh   | Tab Valas   | Wadiah      |               |
| Jawa Barat        | 116,269,854  | 137,169,353 | 286,797,871 | 257,178,886 | 210,249,761 | 151,535,587 | 1,159,201,312 |
| Sumatera Selatan  | 174,798,189  | 99,878,741  | 78,037,371  | 96,772,327  | 118,647,942 | 90,122,712  | 658,257,282   |
| Sulawesi Tenggara |              | 17,667,235  | 230,201,338 | 83,381,153  | 60,919,285  | 56,936,745  | 449,105,756   |
| Jawa Timur        | 31,667,612   | 81,536,259  | 40,132,145  | 11,435,752  |             | 143,760,696 | 308,532,464   |
| Jawa Tengah       | 9,352,014    | 31,735,357  | 77,713,843  |             |             | 114,588,190 | 233,389,404   |
| Sumatera Utara    | 98,558,122   |             | 35,816,640  | 27,343,468  |             | 10,790,087  | 172,508,317   |
| Kalimantan        | 34,429,825   | 12,267,083  | 17,822,099  |             | 4,317,790   | 95,550,495  | 164,387,292   |
| Maluku            |              |             | 5,527,170   | 15,259,978  |             | 128,847,158 | 149,634,306   |
| Bali              | 11,735,440   | 33,602,433  | 89,474,395  |             |             |             | 134,812,268   |
| Sulawesi Utara    | 24,005,422   |             | 8,046,824   |             | 36,574,692  | 53,104,546  | 121,731,484   |
| DKI Jakarta       |              | 7,444,356   | 10,998,310  | 95,853,457  | 5,611,240   |             | 119,907,363   |
| Papua             |              |             | 33,875,897  |             | 50,951,439  | 4,347,030   | 89,174,366    |
| Grand Total       | 500,816,478  | 421,300,817 | 914,443,903 | 587,225,021 | 487,272,149 | 849,583,246 | 3,760,641,614 |





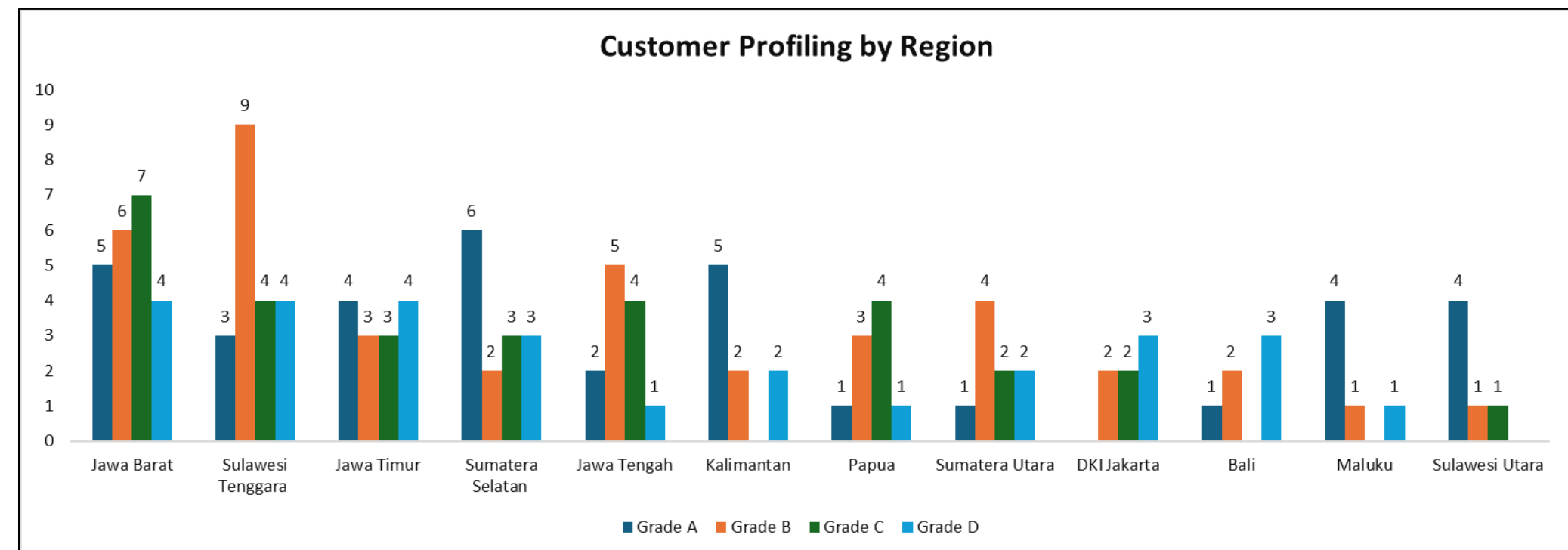
Seara, Menemani Perjalanan  
Datamu Lebih Terarah.

# Lampiran

- Dashboard
- Link Github

## 2. Dashboard Customer Profiling Distribution by Region

| Region            | Customer Profiling |         |         |         | Grand Total |
|-------------------|--------------------|---------|---------|---------|-------------|
|                   | Grade A            | Grade B | Grade C | Grade D |             |
| Jawa Barat        | 5                  | 6       | 7       | 4       | 22          |
| Sulawesi Tenggara | 3                  | 9       | 4       | 4       | 20          |
| Jawa Timur        | 4                  | 3       | 3       | 4       | 14          |
| Sumatera Selatan  | 6                  | 2       | 3       | 3       | 14          |
| Jawa Tengah       | 2                  | 5       | 4       | 1       | 12          |
| Kalimantan        | 5                  | 2       |         | 2       | 9           |
| Papua             | 1                  | 3       | 4       | 1       | 9           |
| Sumatera Utara    | 1                  | 4       | 2       | 2       | 9           |
| DKI Jakarta       |                    | 2       | 2       | 3       | 7           |
| Bali              | 1                  | 2       |         | 3       | 6           |
| Maluku            | 4                  | 1       |         | 1       | 6           |
| Sulawesi Utara    | 4                  | 1       | 1       |         | 6           |
| Grand Total       | 36                 | 40      | 30      | 28      | 134         |





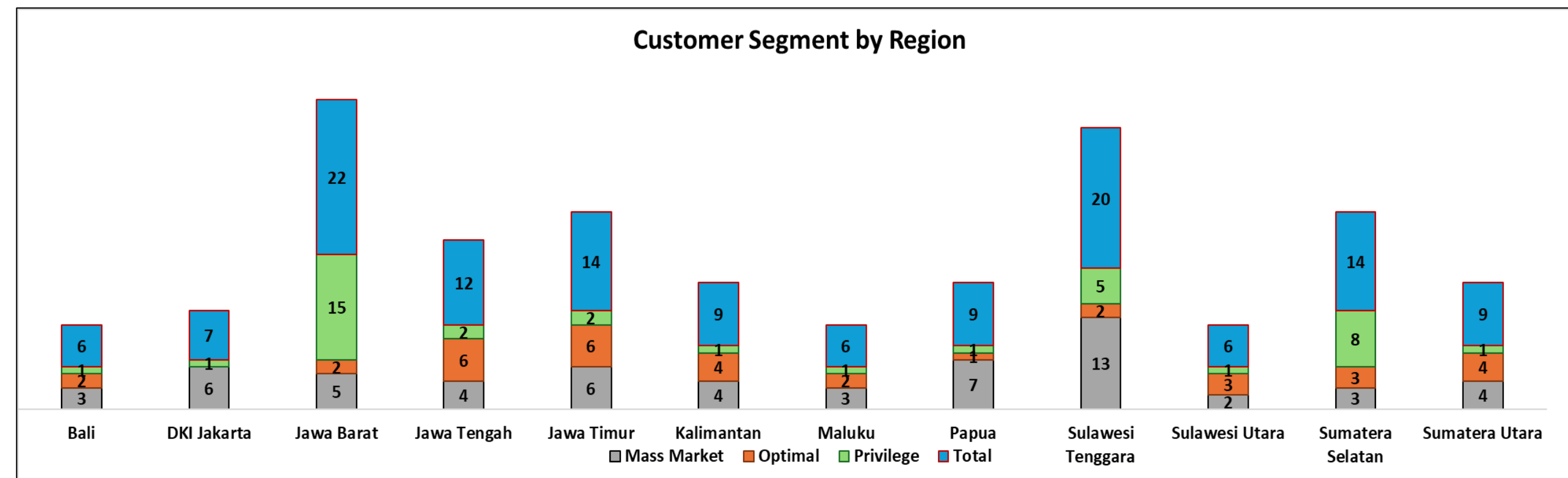
Seara, Menemani Perjalanan  
Datamu Lebih Terarah.

# Lampiran

- Dashboard
- Link Github

## 3. Dashboard Customer Segment Distribution by Region

| Region            | Count of Customer Segmen |         |           |       | Total Balance by Customer Segment |             |               |               |
|-------------------|--------------------------|---------|-----------|-------|-----------------------------------|-------------|---------------|---------------|
|                   | Mass Market              | Optimal | Privilege | Total | Mass Market                       | Optimal     | Privilege     | Total         |
| Bali              | 3                        | 2       | 1         | 6     | 19,054,901                        | 33,602,433  | 82,154,934    | 134,812,268   |
| DKI Jakarta       | 6                        |         | 1         | 7     | 24,053,906                        |             | 95,853,457    | 119,907,363   |
| Jawa Barat        | 5                        | 2       | 15        | 22    | 26,795,227                        | 25,374,028  | 1,107,032,057 | 1,159,201,312 |
| Jawa Tengah       | 4                        | 6       | 2         | 12    | 27,640,270                        | 89,258,247  | 116,490,887   | 233,389,404   |
| Jawa Timur        | 6                        | 6       | 2         | 14    | 33,733,600                        | 94,705,686  | 180,093,178   | 308,532,464   |
| Kalimantan        | 4                        | 4       | 1         | 9     | 6,945,271                         | 61,891,526  | 95,550,495    | 164,387,292   |
| Maluku            | 3                        | 2       | 1         | 6     | 20,787,148                        | 39,630,143  | 89,217,015    | 149,634,306   |
| Papua             | 7                        | 1       | 1         | 9     | 23,213,778                        | 15,009,149  | 50,951,439    | 89,174,366    |
| Sulawesi Tenggara | 13                       | 2       | 5         | 20    | 62,046,686                        | 34,306,675  | 352,752,395   | 449,105,756   |
| Sulawesi Utara    | 2                        | 3       | 1         | 6     | 14,546,680                        | 54,080,258  | 53,104,546    | 121,731,484   |
| Sumatera Selatan  | 3                        | 3       | 8         | 14    | 17,317,509                        | 42,957,833  | 597,981,940   | 658,257,282   |
| Sumatera Utara    | 4                        | 4       | 1         | 9     | 17,717,613                        | 56,232,582  | 98,558,122    | 172,508,317   |
| Grand Total       | 60                       | 35      | 39        | 134   | 293,852,589                       | 547,048,560 | 2,919,740,465 | 3,760,641,614 |





**Seara, Menemani Perjalanan  
Datamu Lebih Terarah.**

# Conclusion

## Key Takeaways

Through this session, I learned the end-to-end data analyst workflow. On Day 1, I focused on data analytics fundamentals and Data Cleansing using text functions (**TRIM**, **LEFT/MID/RIGHT**, **SUBSTITUTE**, **LEN**) to standardize raw data. On Day 2, I advanced to Data Processing & Joining using conditional logic (**IF**, **IFS**) and dynamic lookups (**XLOOKUP**, **INDEX-MATCH**). Finally, I performed Data Aggregation using conditional calculations (**SUMIF**, **COUNTIF**, **AVERAGEIF**) to summarize large datasets into clear visualizations and strategic business insights.



**Seara, Menemani Perjalanan  
Datamu Lebih Terarah.**



# Let's Connect



 **LinkedIn : Rahmanida Susiana**

 **Email : [Rahmanidasusiana@gmail.com](mailto:Rahmanidasusiana@gmail.com)**

 **GitHub : <https://github.com/Rahmanida01>**

